

Youssef Helmy

Alexandria, Egypt

Phone: +20 112 223 7213

Email: youssefhelmy4444@gmail.com

LinkedIn: linkedin.com/in/youssefhelmyegy

GitHub: github.com/YoussefMoHlemyAlpha

Military Status: Exempted

PROFESSIONAL SUMMARY

AI Engineer specializing in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Backend Development, and Intelligent Systems. Experienced in designing scalable AI applications using FastAPI, Node.js, TensorFlow, LangChain, FAISS, Neo4j, and modern LLM frameworks. Completed 3 technical internships and 4 professional certifications in Machine Learning and Deep Learning. Passionate about building production-ready AI systems that combine machine learning, backend engineering, and real-time data processing.

EDUCATION

Alexandria University

B.Sc. in Intelligent Systems, Faculty of Computers and Data Science | GPA: 3.91

Alexandria, Egypt

Aug 2022 – May 2026

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, SQL, C/C++, Java

AI / Machine Learning: TensorFlow, Keras, Scikit-learn, LangChain, FAISS, OpenCV, NLP, CNN, RAG

Backend Development: FastAPI, Node.js, Express.js, NestJS, REST APIs

Databases: MongoDB, MySQL, Firebase Realtime Database, Neo4j

Cloud & DevOps: Docker, AWS S3, Git, GitHub, Postman, WebSocket

Data Science: NumPy, Pandas, Matplotlib, Seaborn, Streamlit

IoT: ESP32, MQTT, HiveMQ, Embedded C/C++

Soft Skills: Problem-Solving, Team Collaboration, Communication, Adaptability, Critical Thinking, Time Management

EXPERIENCE

Front End Intern

July 2025 – August 2025

Information Technology Institute (ITI)

Alexandria, Egypt

- Developed responsive web interfaces using semantic HTML5, CSS3, and modern JavaScript (ES6+), improving maintainability through modular code architecture; **achieved a perfect 100% score** on all assessed deliverables.
- Implemented modern JavaScript features including arrow functions, template literals, and modular code architecture.

Artificial Intelligence Intern

August 2024 – September 2024

Telecom Egypt

Alexandria, Egypt

- Developed ML models (Linear and Logistic Regression) using TensorFlow, Keras, Scikit-Learn, NumPy, and Pandas, **achieving 96% model accuracy**.
- Collaborated on Computer Vision and NLP projects using deep learning techniques for image analysis and text processing.

IoT Systems Intern

July 2024 – September 2024

IEEE AlexCSC

Alexandria, Egypt

- Developed ESP32-based IoT applications using C/C++ with MQTT protocols and Firebase for real-time data synchronization; **awarded Grade A** for overall performance.
- Gained expertise in network protocols, cloud computing, and IoT communication patterns for multi-device applications.

PROJECTS

SEBAYT LMS Builder

[GitHub](#)

Graduation Project, Alexandria University — Grade: A

2025 – 2026

Python • Neo4j • PyMuPDF • Gemma 3 • Llama 3.2 • Qwen3-VL • Distil-Whisper • FAISS • Ollama • CUDA

- Built a multi-tenant SaaS platform enabling non-technical users to create a fully branded LMS with custom themes and integrated payment gateways, with no coding required.
- **Quiz RAG Generation:** Built a Retrieval-Augmented Generation pipeline using **Neo4j**, **PyMuPDF**, and **Gemma 3** to construct knowledge graphs from PDF content, ranking and selecting the top 15 highest-value entities per lesson.
- Implemented **BFS** graph traversal for context retrieval and a Generator-Critic architecture combining **Gemma 3** and **Llama 3.2**. Enforced a minimum 50% keyword-coverage threshold and automatically rejected any generated question with a hallucination rate above 30%.
- **Video RAG:** Combined **Distil-Whisper** speech-to-text and **Qwen3-VL** visual captioning into a structured, timestamped JSON index segmented into 30-second clips for multimodal video understanding.
- Built a **FAISS**-based semantic search layer with **all-MiniLM-L6-v2** embeddings and a 4-billion-parameter **Gemma 3** model on **Ollama/CUDA**, achieving sub-second query response times.
- **Bloom's Taxonomy Classifier:** Built a bilingual (Arabic/English) question-classification system using In-Context Learning and RAG with **Qwen3** on **Ollama**, requiring only 2 to 4 few-shot examples per cognitive level.
- Implemented PDF extraction, Arabic text normalization, few-shot prompting, and fuzzy-match post-processing — eliminating the need for labeled training data.
- Collaborated within a 5-member team under faculty supervision and delivered complete technical documentation; **awarded Grade A**.

Chat with Multiple PDFs

[GitHub](#)

Personal Project

2025

Python • Streamlit • LangChain • Google Gemini • FAISS • PyPDF2

- Built a RAG-based web app using **PyPDF2** for text extraction and **Google Gemini** embeddings.
- Implemented a **FAISS** vector store for retrieval and orchestrated the QA pipeline with **LangChain**.
- Deployed an interactive **Streamlit** interface for real-time multi-document upload and querying.

Medical ChatBot

[GitHub](#)

Personal Project

2025

FastAPI • Ollama • FAISS • Sentence Transformers • HTML • CSS • JavaScript

- Built a full-stack medical chatbot with a **FastAPI** backend and **Ollama (Llama 2)** LLM, covering 7+ medical conditions including diabetes, cancer, and influenza.
- Implemented retrieval using **FAISS** and **sentence-transformers** embeddings, sourcing answers from **MedlinePlus**.
- Developed a responsive HTML/CSS/JS frontend with real-time feedback, directing users to licensed healthcare providers for personal advice.

CERTIFICATIONS

Backend Node.js Diploma

December 2025

Route Academy

Machine Learning Specialization

June 2024

Coursera – Andrew Ng

Applied Deep Learning

July 2024

Mahara Tech

Python for Data Science

June 2024

IBM SkillsBuild